

OPENMANET · FIELD EXTENSION

FIELD EXTENSION FOR OPENMANET

The Complete Guide

Release v1.0

Node package 1.0.9 · ATAK plugin ManetNode FE 1.4 · ManetNode app 8.3

OpenMANET 1.8 · Raspberry Pi 4 · HaLow · LoRa (Meshtastic)

CONTENTS

1 What it is and how it fits together
2 Parts and assembly
3 Setting up nodes
4 The status light
5 Power and battery
6 The node web page
7 The ATAK plugin (ManetNode FE)
8 The ManetNode app
9 Map tools: points, geofences, breadcrumbs
10 Capabilities Test (range walk)
11 LoRa fallback
12 The mesh: HaLow and 2.4 GHz
13 Maintenance, updates and backups
14 Troubleshooting
15 Reference: ports, files, API
16 Building from source

HOW TO READ THIS GUIDE

Chapters 1–4 are what every team member should know. Chapters 5–13 explain each part in depth. Chapters 14–16 are for whoever looks after the nodes. Words in **BOXES** are buttons or tabs you press.

DOWNLOADS

Every file is attached to the **v1.0 release** on the GitHub page (Releases, right-hand side):

…SD-image-rpi4.img.gz	Ready-to-flash SD card for a new node (OpenMANET 1.8 + Field Extension)
field-extensions_1.0.9_all.ipk	Node package, to install on an existing OpenMANET 1.8 node
…ATAK-plugin-ATAK-5.7.0.apk / 5.8.0.apk	ATAK plugin, signed by TAK.gov, for regular ATAK CIV 5.7 / 5.8
…ManetNode-Android.apk	ManetNode phone app (Android)
…ManetNode-Mac.zip	Mac launcher that lists the nodes
…field-card.pdf	One-page card to keep with each node
SHA256SUMS.txt	Checksums to confirm every download is intact

1 · WHAT IT IS AND HOW IT FITS TOGETHER

OpenMANET turns a Raspberry Pi 4 with a Wi-Fi HaLow radio into a mesh node: nodes find each other and form one network over long range, and phones join a node's Wi-Fi to get on it. The **Field Extension** adds what a team needs in the field:

- **A status light** on each node that tells you at a glance whether it has mesh, internet, LoRa, GPS, how charged it is, and when it is being worked on.
- **Battery and power** control for the PiSugar battery: percent and time left, auto power-on, safe low-battery shutdown, clean shutdown on a long press.
- **LoRa fallback**: when the mesh is out of reach, ATAK positions, chat and map markers go over a Meshtastic LoRa radio instead.
- **Self-repair**: watchdogs restart a stuck HaLow radio, demote a bad 2.4 GHz link, recover after crashes.
- **A node web page** (port 8787), the **ATAK plugin** and the **ManetNode app** to see and change all of it.

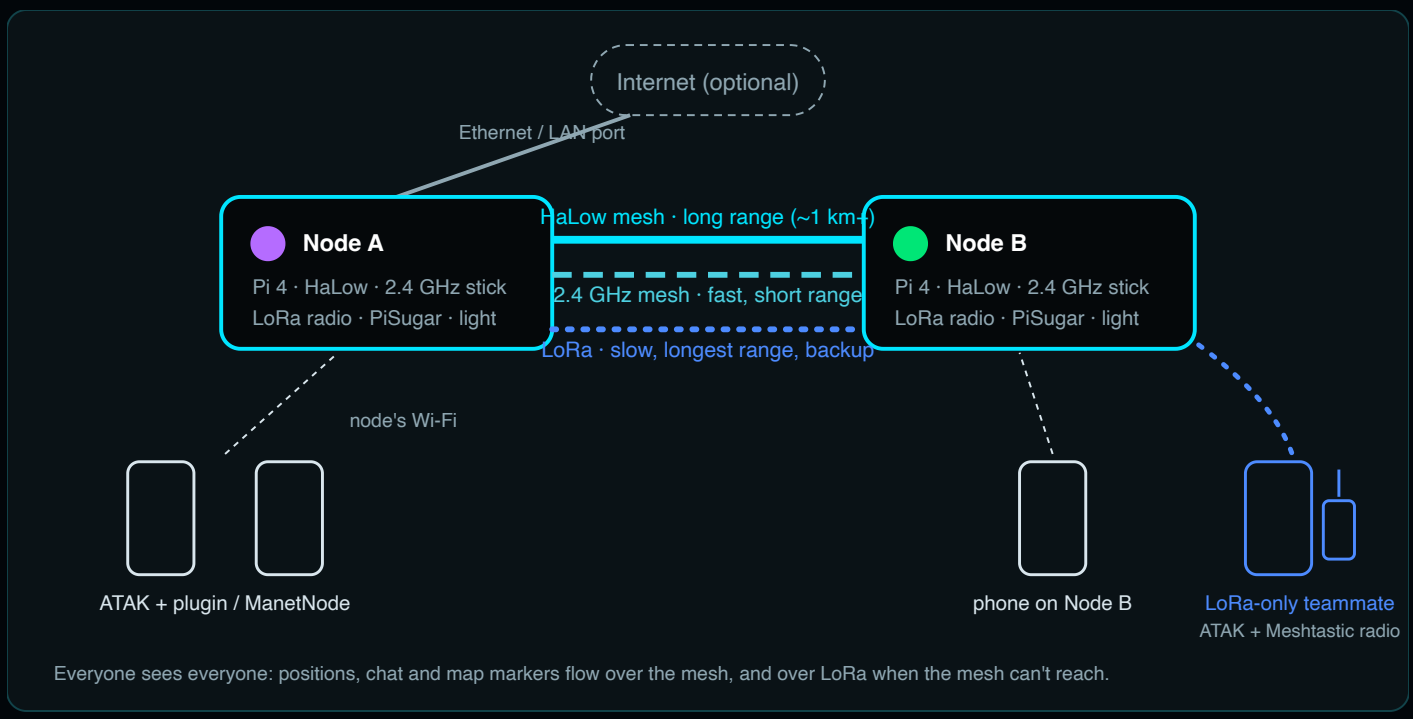


Figure 1. Two nodes, phones on each node's Wi-Fi, and a teammate who only has a LoRa radio.

The three ways to control it

TOOL	WHERE	BEST FOR
ATAK plugin "ManetNode FE"	Inside ATAK (Tools menu)	Day-to-day: light, battery, LoRa, push-to-talk, without leaving ATAK
ManetNode app	Android app	Team map with points, geofences and breadcrumbs, chat, PTT, every node setting, adding nodes, the Capabilities Test
Node web page	Browser: `http://<node>:8787`	Everything in depth, from any phone or laptop, no install

OpenMANET's own page stays at port 8080 (mesh, Wi-Fi and system settings). The Field Extension never changes OpenMANET itself.

2 · PARTS AND ASSEMBLY

PART	NEEDED?	NOTES
Raspberry Pi 4 (4 or 8 GB)	Yes	The node computer.
MM6108 HaLow HAT (SPI) + antenna	Yes	The long-range mesh radio OpenMANET is built for.
microSD card, 16 GB or more	Yes	Holds OpenMANET + Field Extension.
PiSugar 3 / 3 Plus battery	Recommended	Battery %, time left, auto power-on, clean shutdown.
RGB LED (common cathode) + three 330 Ω resistors	Recommended	The status light. Wiring in Figure 3.
USB Meshtastic radio (e.g. Heltec V4)	Recommended	LoRa fallback. Any Meshtastic radio on USB works.
USB 2.4 GHz Wi-Fi stick (RT5370)	Optional	A second, faster mesh link between nearby nodes.
USB GPS	Optional	Places the node on maps. Without it, a phone can give the node its position.

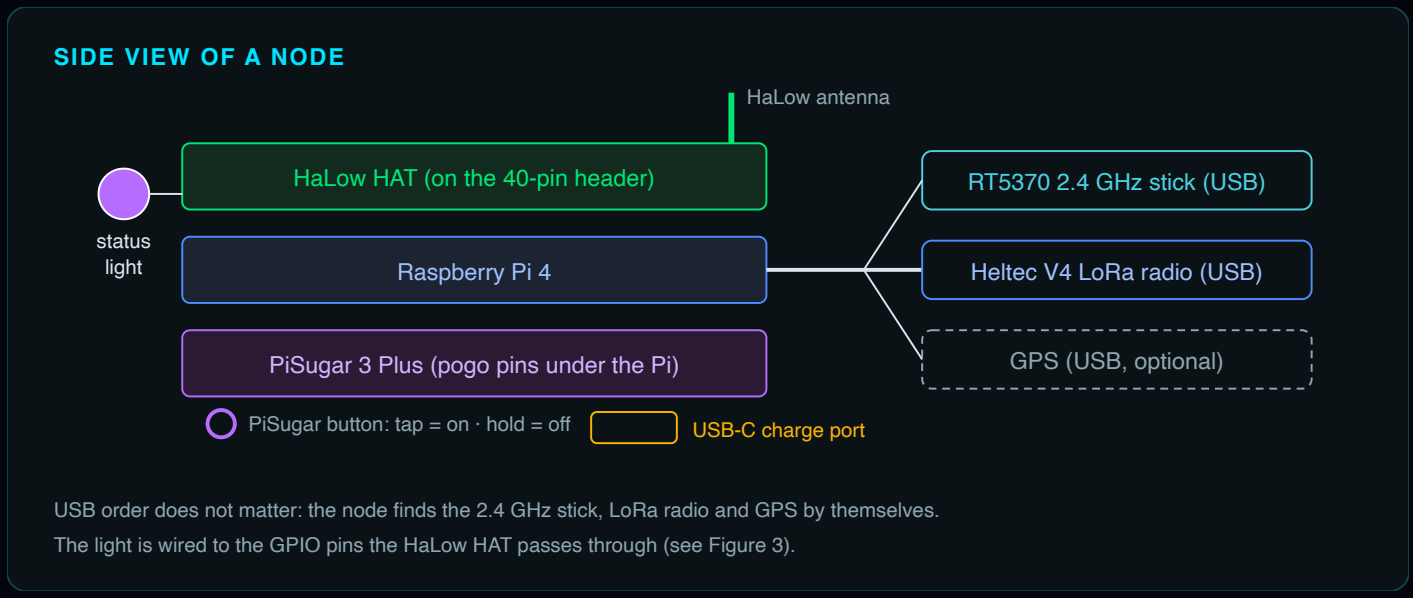


Figure 2. What goes where.

2.1 Wiring the status light

Use a **common-cathode** RGB LED: one long leg goes to ground, the other three legs are red, green and blue. Put a **330 Ω resistor** in each colour leg. The three colour pins are fixed in the software: **red = GPIO 27, green = GPIO 22, blue = GPIO 26**.

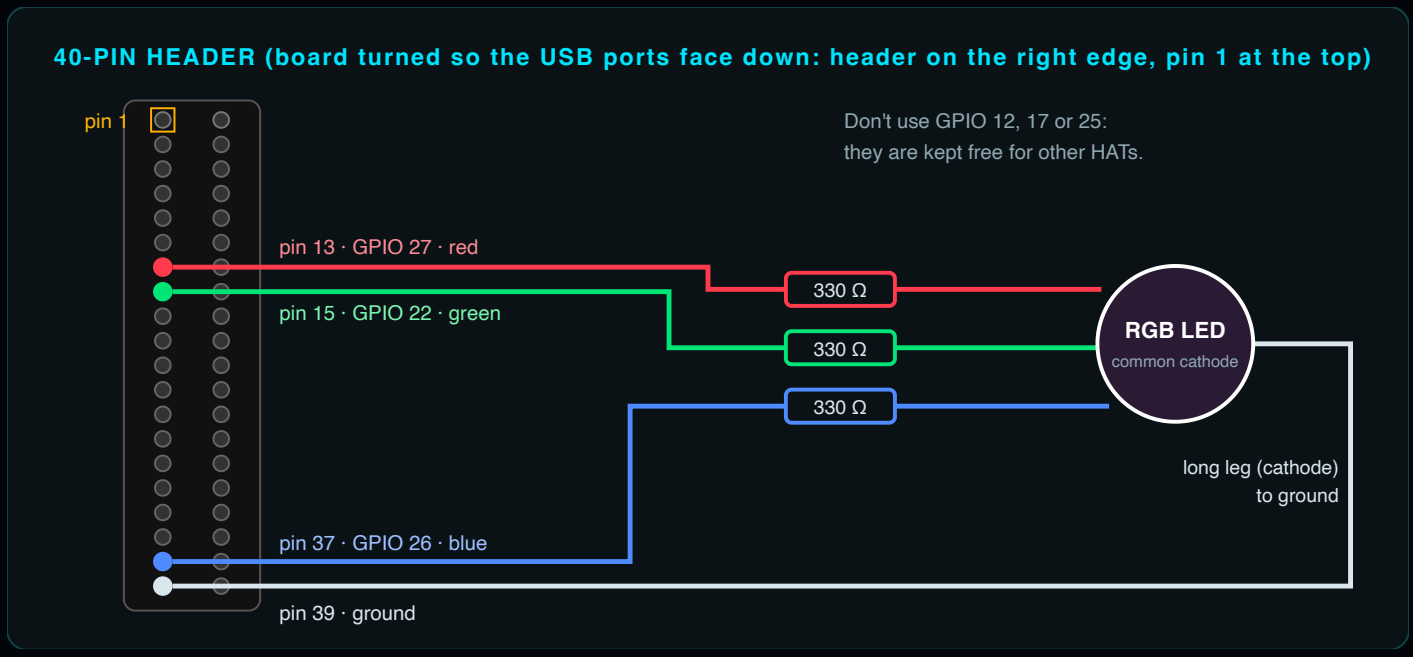


Figure 3. Status light wiring. The node checks the pins at start-up and refuses a pin that clashes with something else.

CHECK IT

After the first boot the light should go **red** (starting), then turn the colour of the node's state. The self-test (chapter 13) should say “PASS Status light driver is running”.

2.2 Charging through a case USB-C port

Many cases bring the PiSugar's charge input out to a USB-C socket on the case. A plain USB-C socket gives **0 V** on a USB-C-to-C charger unless its two **CC** pins each have a **5.1 kΩ** resistor to ground; that is how the charger knows a device is plugged in. A USB-A-to-C cable always works, because it does not need the CC pins.

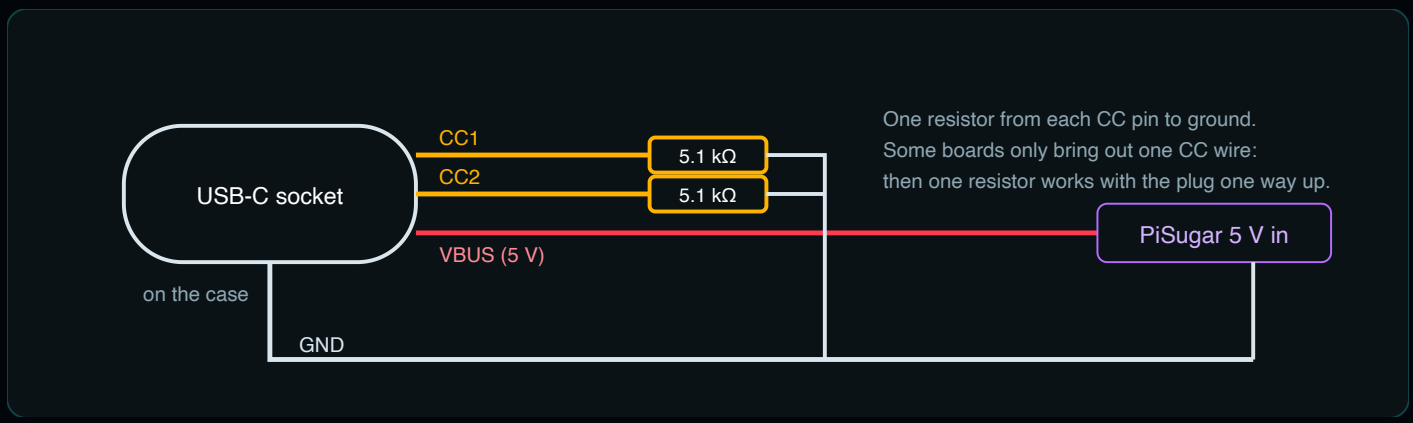


Figure 4. The two 5.1 kΩ “CC” resistors that make a USB-C charger switch on.

CHARGER

Use a 5 V / 3 A supply and a short cable. A weak supply can't run the node and charge the battery at the same time, and the battery slowly drains even while plugged in.

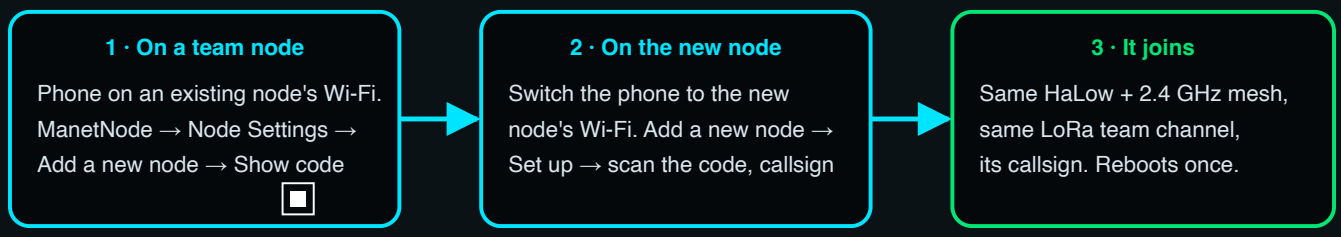
3 · SETTING UP NODES

3.1 A brand-new node from the SD card image

1. Download `FieldExtension-OpenMANET-v1.0-SD-image-rpi4.img.gz` from the release.
2. Flash it to a microSD card with **Raspberry Pi Imager** (CHOOSE OS → USE CUSTOM) or **balenaEtcher**. Both accept the .gz file directly.
3. Put the card in the Pi and power it on. The light goes red while it starts (about a minute).
4. Join the new node's Wi-Fi and finish **OpenMANET's own first-boot setup** on its page (port 8080): root password and country/region.
5. Add it to your team with a **setup code** (next section). That gives it the same mesh, the same LoRa team channel and its own callsign in one step.

The image is stock OpenMANET 1.8 (Pi 4, HaLow MM6108 SPI) with the Field Extension and its Python preinstalled. It holds no keys, passwords, callsigns or mesh settings. It also carries the optional PortableTAK loader, which does nothing unless a signed PortableTAK drive is plugged in.

3.2 Adding a node to the team with a setup code



The code is also on the web page (Maintenance → Add a new node): Show setup code on one node, paste it into "Paste a setup code" on the new one. It contains the mesh password and the team key: only show it to your team.

Figure 5. One QR code sets up a new node.

After the reboot (about 2 minutes) the new node's light turns **green** or **purple** as soon as it hears the others, and it appears in everyone's node list and map.

3.3 Installing on an existing OpenMANET 1.8 node

1. Put `field-extensions_1.0.9_all.ipk` into the `node-installer/` folder (from the GitHub page).
2. Join the node's Wi-Fi (or plug into its LAN port).
3. Mac: double-click `Install Field Extension (Mac).command` (first time: right-click → Open). Windows: right-click the `.ps1` → Run with PowerShell.
4. Press Return for 10.41.0.1 (or type the node's address) and type the node's root password when asked.

It saves a backup of the node's settings next to the installer, installs the package, and runs the self-test. Stock OpenMANET has no Python, so the first install fetches it from the OpenWrt package feed: **the node needs internet** for that (plug its LAN port into a router).

3.4 Phones

WHAT	HOW
ATAK plugin	Install regular ATAK CIV 5.7 or 5.8. Copy the matching plugin APK to the phone and tap it. In ATAK, load **ManetNode FE** from the Plugins list. It finds the node by itself whenever the phone is on a node's Wi-Fi.
ManetNode app	Copy the Android APK to the phone and tap it (allow “Install unknown apps” for your file manager). Allow location and microphone when asked.
Any phone or laptop	Join a node's Wi-Fi and open `http://10.41.0.1:8787` (or the node's own address).

> **NODE PASSWORD**
> Anything that changes a node asks for the node's password once: it is the same root password as the OpenMANET page. The app and plugin keep it encrypted on the phone (Android Keystore), so you only type it once.

4 · THE STATUS LIGHT

The light shows **one state at a time**, the most important one. Short flashes for GPS and charging ride on top of it. Every pattern repeats, so watch for about 10 seconds.

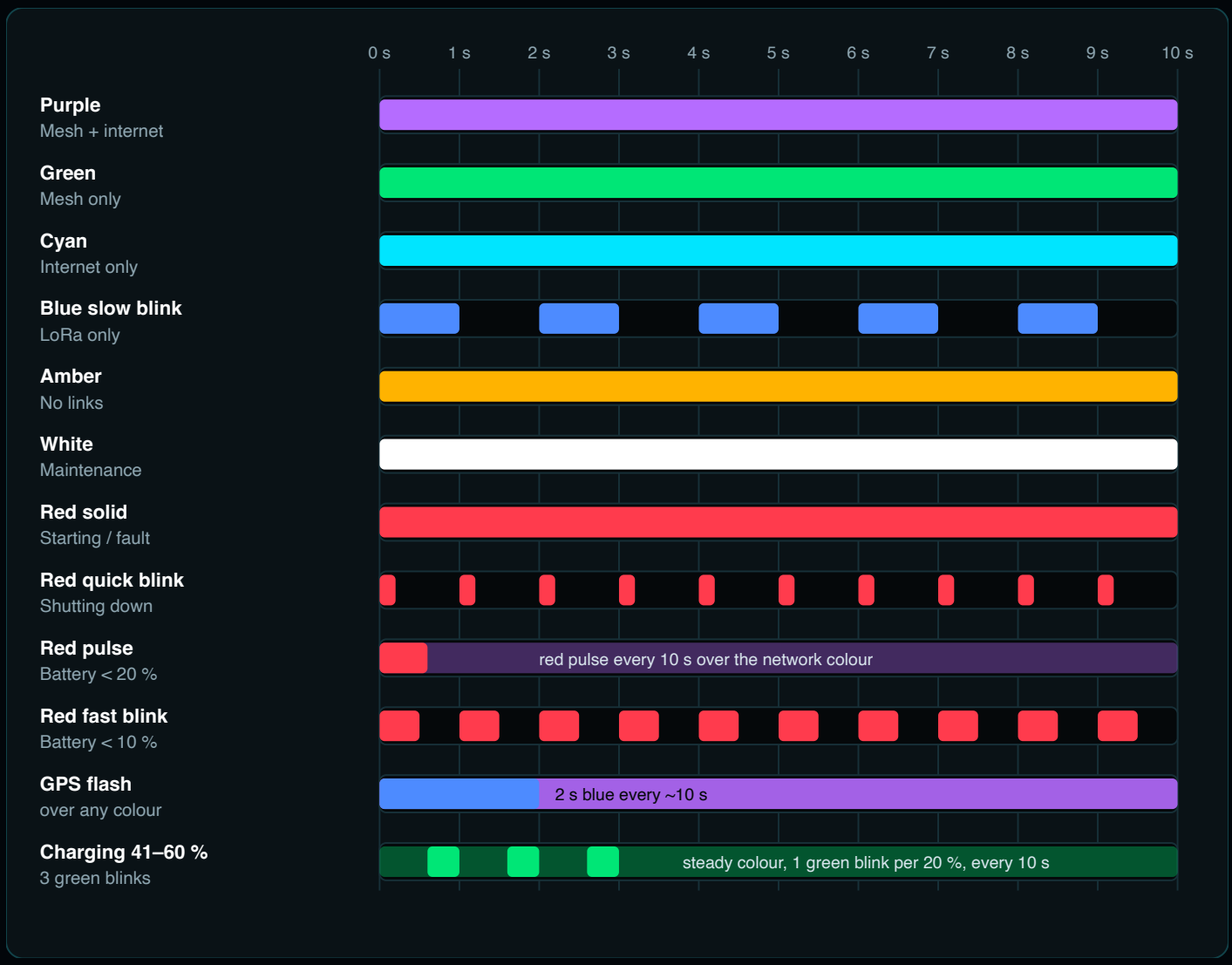


Figure 6. Every light pattern over 10 seconds. Filled = lit.

4.1 Which state wins

When several things are true at once, the higher one in this list is shown:

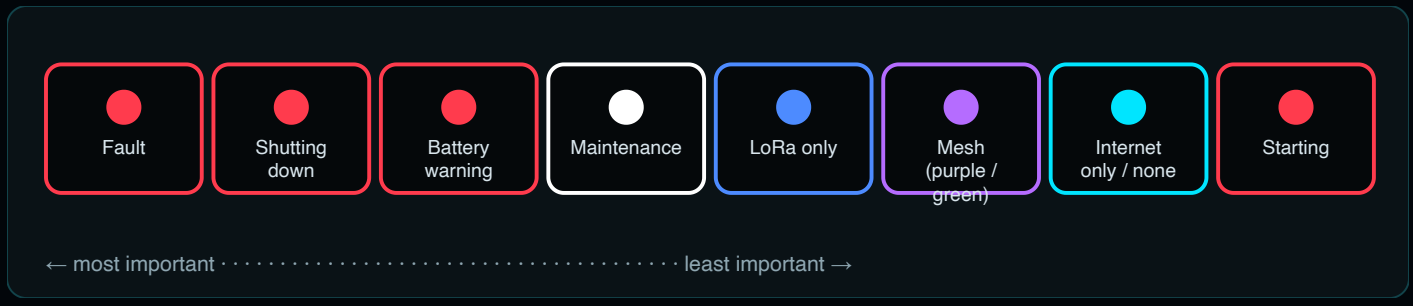


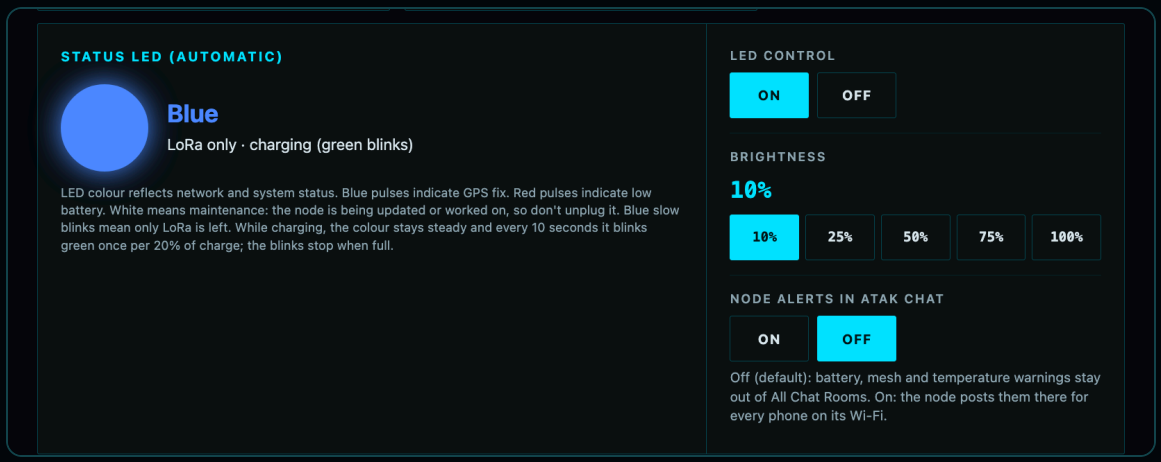
Figure 7. Priority: a fault beats everything; a low battery beats maintenance; maintenance beats the normal network colours.

4.2 What each light means and what to do

LIGHT	MEANS	DO
Purple	Mesh + internet	All good.
Green	Mesh, no internet	Normal in the field.
Cyan	Internet but no other node heard, and no LoRa radio	Normal for a single node on a network cable.
Blue slow blink	LoRa only: no other node on the mesh; the LoRa radio carries positions and chat	Move closer to the team or raise the node to get the mesh back.
Amber	Nothing: no mesh, no internet, no LoRa	Check the other nodes are on; reseat the LoRa radio.
White	Maintenance: updating, applying settings, running diagnostics, or Maintenance mode switched on	**Don't unplug.** It clears by itself; manual mode ends after 30 min.
Red solid	Starting up (first minute), or a fault	If it stays red, open the web page: Node Health shows what failed.
Red quick blink	Shutting down cleanly	Wait until it goes dark before unplugging.
Red pulse	Battery under 20 % (on battery)	Charge soon.
Red fast blink	Battery under 10 %	Charge now; the node shuts itself down safely at 5 %.
Blue 2 s flash	GPS has a fix (every ~10 s)	—
Green blinks	Charging: the colour stays steady and every 10 s it blinks green once per 20 % of charge	5 blinks = nearly full. The blinks stop when it's full.

DIM IT OR TURN IT OFF

Web page **STATUS LED**, plugin **SET** tab, or ManetNode Node Settings. Brightness 10–100 %. Off is useful at night; the node keeps working.



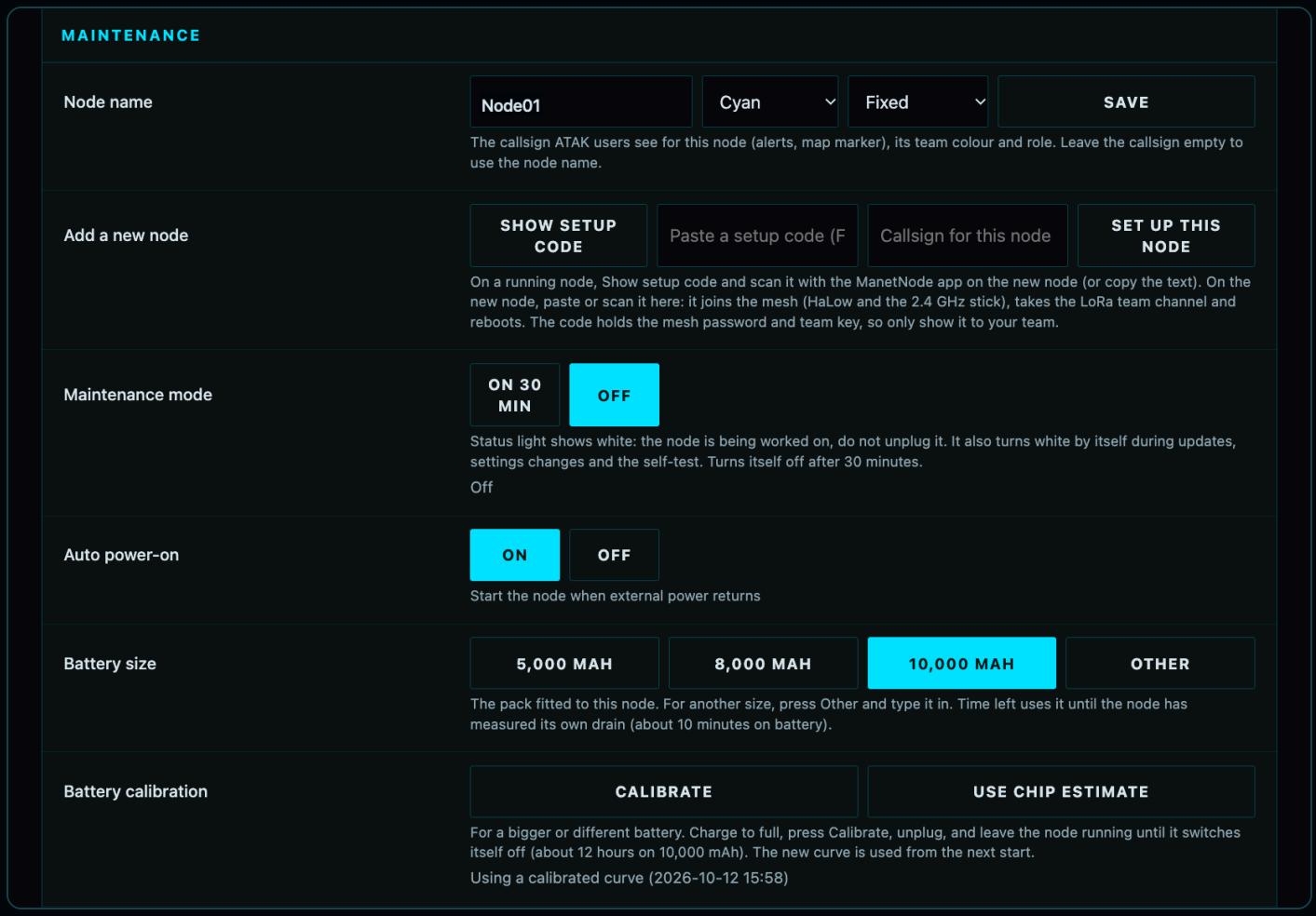
The light's state, colour and meaning are also shown live on the web page, the plugin and the app.

5 · POWER AND BATTERY

FEATURE	WHAT IT DOES
Battery %	From the PiSugar, smoothed over 5 minutes so it doesn't jump. After a calibration it uses the node's own measured curve. While charging the % rises slowly on purpose; the voltage is the honest number.
Time left	From the pack size you set (5,000 / 8,000 / 10,000 mAh or other) and, after about 10 minutes on battery, the node's real drain.
Auto power-on	ON: after the battery ran flat, the node starts by itself as soon as the charger is plugged in. Recommended ON for every node.
Safe shutdown	On battery, at 5 % (or 10 %) the node shuts down cleanly instead of losing power suddenly, which can damage the SD card.
Button long press	**Clean shutdown** (recommended): hold the PiSugar button; the light blinks red and the node saves everything and powers off. **Cut power**: the PiSugar default, off at once.
Calibration	For a new or different battery: charge to full, press **CALIBRATE**, unplug, and leave the node running until it switches itself off (about 12 h on 10,000 mAh). Plugging in cancels the run.

TURNING A NODE ON AND OFF

On: short-press (tap) the PiSugar button. **Off:** hold the button until the light blinks red, or use **SHUTDOWN** in the plugin, app or web page. It stays off until the button is pressed again (or the charger is plugged in after a full drain, with auto power-on ON).



Web page → Maintenance: node name, adding a node, auto power-on, battery size, calibration and safe shutdown.

6 · THE NODE WEB PAGE

Open `http://<node address>:8787` on any device on the node's Wi-Fi (the first node is usually `10.41.0.1`). The top of the page shows the node's name, uptime, the tiles, and buttons for the **NODES** list and the **OPENMANET** page.



The status tiles: battery, GPS, mesh neighbours, internet, mesh IP, HaLow radio, the phone Wi-Fi and the backhaul.

TILE	SHOWS
B · Battery	%, voltage, charging or on battery, time left, auto power-on, calibration.
G · GPS	Fix or no fix, and where the node's position comes from.
M · Mesh Neighbours	How many nodes it hears, and the speed of each link (HaLow and 2.4 GHz).
I · Internet	Online when a real check to the internet passes, not just when a route exists.
RF · HaLow Radio	Frequency and bandwidth.
AP · EUD Wi-Fi	The Wi-Fi name phones join.
BH · Backhaul	How the node reaches the internet (Ethernet, mesh gateway, none).

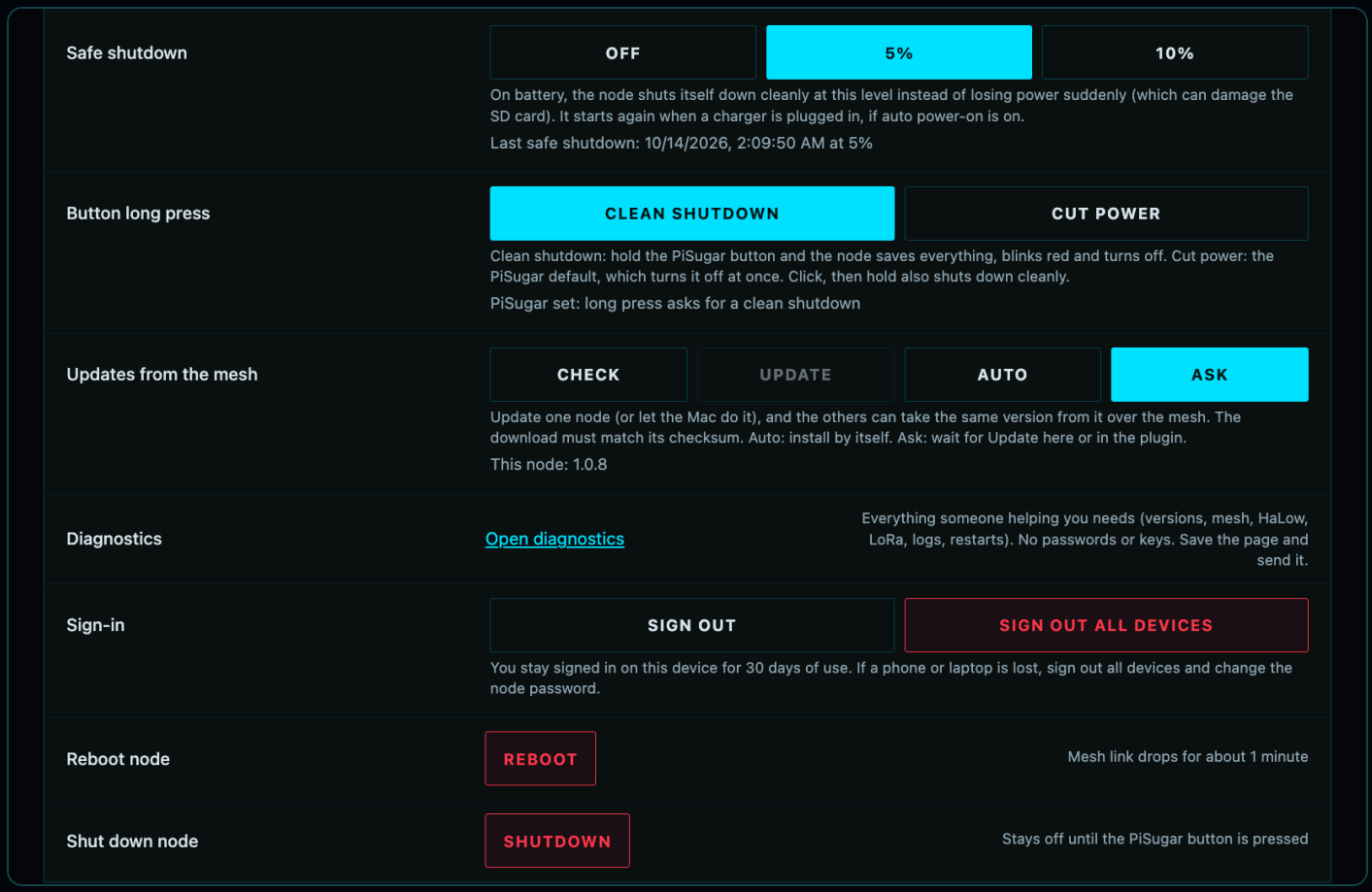
SERVICES		
openmanetd	● Running	OpenMANET daemon and web UI
gpsd	● Running	GPS receiver service
fe-web	● Running	Field Extension web page and API
fe-light	● Running	Status light and PiSugar button
fe-phone-internet	● Running	Phone internet helper
fe-lora	● Running	LoRa fallback and radio control
fe-recovery	● Running	Self-recovery service
fe-halow-watchdog	● Running	Reboots if the HaLow radio stops (at most 2 in 6 h)
fe-peer-watchdog	● Running	Reboots if HaLow peers vanish for 20 min (at most 2 in 6 h)
fe-rate-latch	● Running	Unsticks a stuck HaLow rate
fe-link-check	● Running	Checks each mesh link
fe-link-demote	● Running	Moves traffic off a weak 2.4 GHz link
fe-mesh-guard	● Running	Keeps mesh forwarding off (batman does the routing)
fe-traffic-shaping	● Off	Traffic shaping (off until tested)
fe-power-off	● Running	Turns the node fully off after a clean shutdown

Services: every part of the Field Extension and OpenMANET, with what it does. All should say Running (traffic shaping is off unless limited).

NODE HEALTH		
CPU Temperature	● 74.5 °C	Processor thermal state
CPU Load	● 0.21	1 minute load average
Memory	● 15%	1160 of 7818 MB in use
HaLow Interface	● ONLINE	wlh0 · 914.0 MHz
Kernel	6.6.138	aarch64
HaLow watchdog	● On	HaLow radio answering
HaLow peer watchdog	● On	No HaLow peer yet this boot (nothing to recover)
HaLow rate check	● On	HaLow width unknown; not checking
Mesh guard	● OK	Mesh forwarding off
Last restart	● Unplanned	2026-10-26 18:23 unexpected (power loss or crash) up 12391s

Node Health: temperature, load, memory, HaLow, watchdogs, mesh links and how the last restart ended.

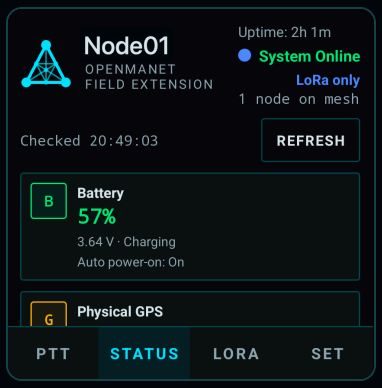
Further down: **Backhaul**, **Phone Internet** (chapter 12), **LoRa** (chapter 11), **Maintenance** (chapters 5 and 13) and **OpenMANET Updates**.



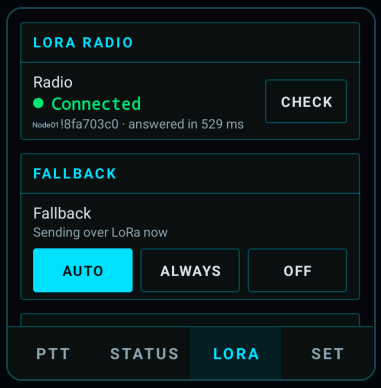
Maintenance, continued: long press, updates from the mesh, diagnostics, sign-in and reboot.

7 · THE ATAK PLUGIN (MANETNODE FE)

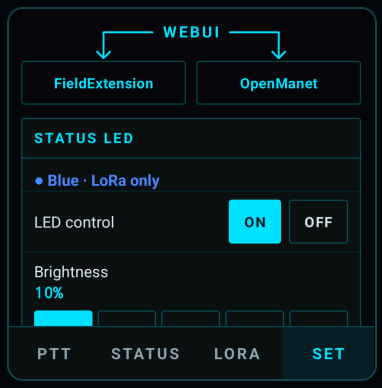
Open ATAK, tap the Tools menu, then **ManetNode FE**. The plugin finds the node your phone's Wi-Fi is on. To point it at another node, long-press the node name and type its address. The plugin's corner dot on the map is the node's light colour.



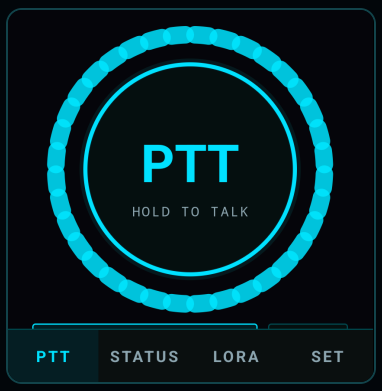
STATUS: the node, its light state and battery



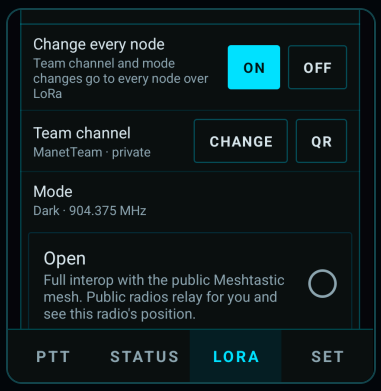
LORA: radio check and fallback



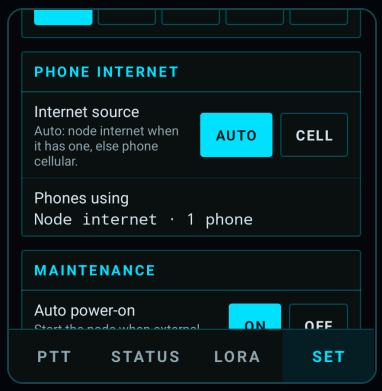
SET: web page buttons and the status light



PTT: hold to talk



LORA, further down: team channel and mode



SET, further down: phone internet and maintenance

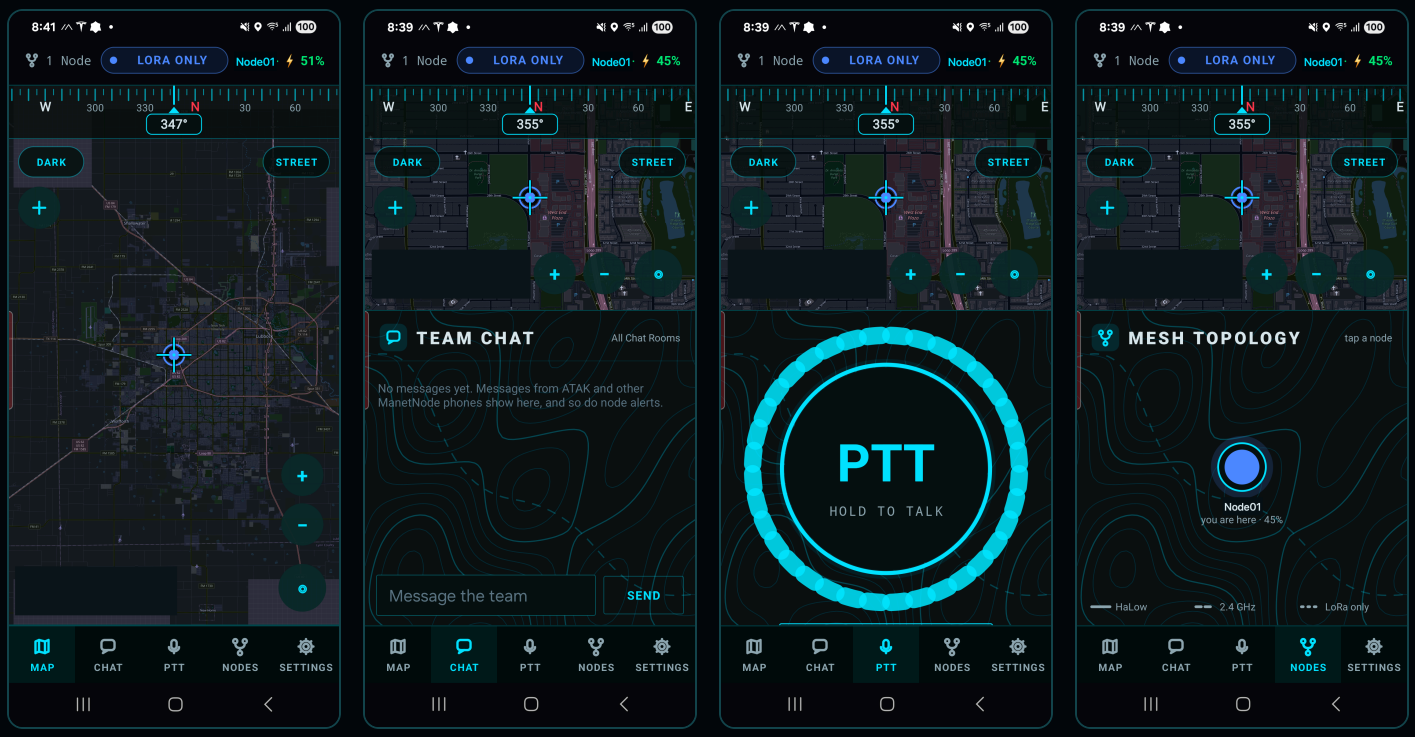
TAB	WHAT'S ON IT
PTT	Push-to-talk to the team over the mesh, with talk groups. Same voice channel as the ManetNode app.
STATUS	The light and its meaning, battery, mesh links, internet, LoRa, services and node health.
LORA	Fallback mode, LoRa mode (Open / Covert / Dark), team channel with QR code, radios heard, send a test message.
SET	Callsign, team and role; light on/off and brightness; auto power-on and long press; position source; node markers and mesh lines on the ATAK map; clear fleet chat; new-node setup code; Maintenance mode; reboot and shutdown. A button opens the node's full web page.

NODES ON THE ATAK MAP

With SET → Nodes on map ON, every node shows as a marker in its light colour, with lines between nodes: solid = HaLow, dashed = 2.4 GHz, dotted = LoRa only.

8 · THE MANETNODE APP

ManetNode is a field app for Android: a status bar on top, the team map, and five tabs at the bottom.

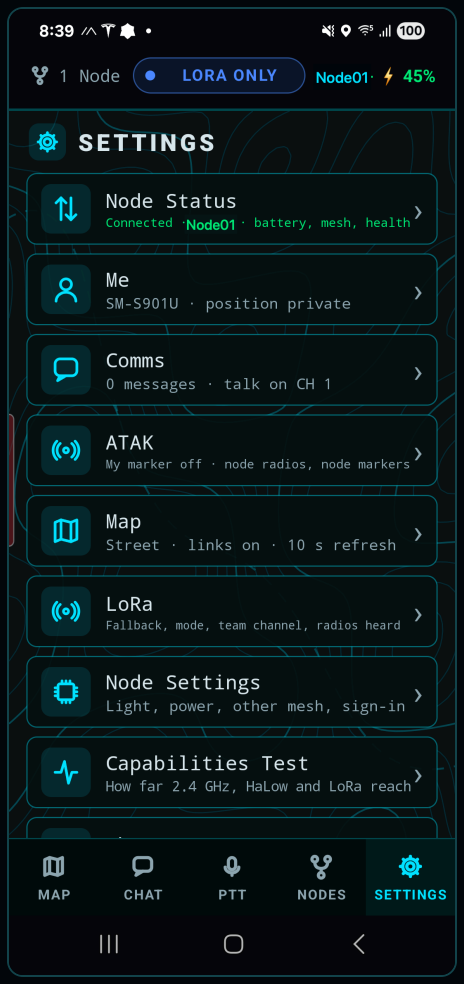


MAP | CHAT | PTT | NODES (mesh topology)

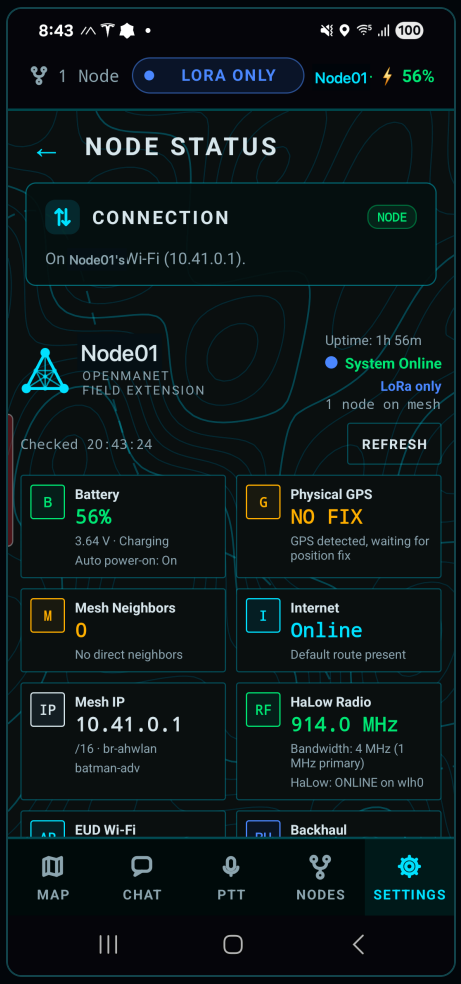
Location details are blacked out in these pictures.

TAB	WHAT IT DOES
MAP	Nodes (coloured like their light), team members, mesh links, your position, compass tape, crosshair with coordinates, MGRS and the street address. Points, geofences and breadcrumbs (chapter 9).
CHAT	ATAK's All Chat Rooms: ATAK users and app users talk together; over LoRa when the mesh is down.
PTT	Push-to-talk with talk groups 1–5, same channel as the plugin.
NODES	Mesh topology: every node and how they connect.
SETTINGS	Node Status, Me (callsign, position sharing, ATAK sharing), Comms, ATAK, Map, LoRa, Node Settings, Capabilities Test, About.

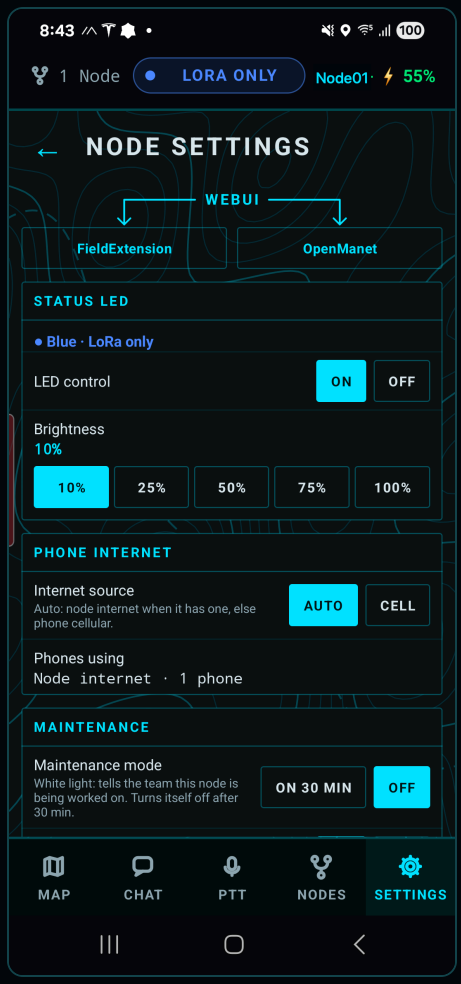
The top bar shows how many nodes are on the mesh, the connection state (mesh, LoRa only, standalone) and the node your phone is on with its battery.



SETTINGS menu



Node Status



Node Settings, with Maintenance mode

NODE SETTINGS IN THE APP

Everything from the web page's Maintenance section: light, power, battery, position source, joining another mesh, adding a new node, Maintenance mode, reboot and shutdown.

9 · MAP TOOLS: POINTS, GEOFENCES, BREADCRUMBS

Tap the round (+) button on the left of the map for the Map Tools menu, or long-press anywhere on the map to drop a point there.



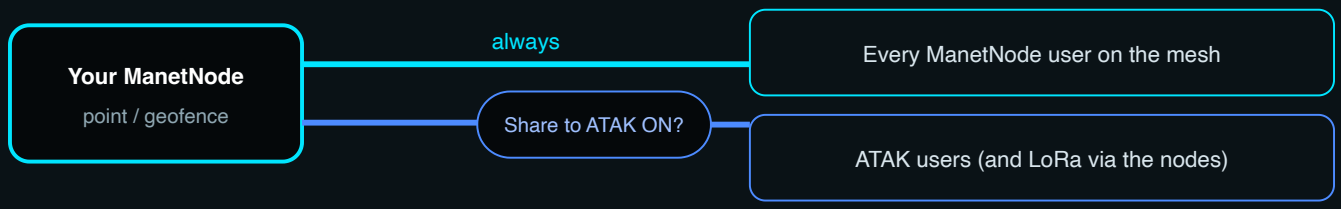
The Map Tools menu

Points and geofences are stored on the phone and shared over the mesh, so they survive an app restart and show up for teammates who join later (each phone re-sends its own items every 2 minutes).

Tap any point or fence for its details. Long-press the map for a quick point at that spot.

TOOL	HOW
Drop a point	Name it and pick a type: Waypoint, Rally point, Friendly, Hostile, Unknown, Neutral. Notes are optional. Tap a point later for who placed it, distance and bearing from you, coordinates and MGRS, and Edit / Delete.
Geofence circle	Centred on the crosshair; set the radius in metres.
Geofence area	Tap the map to place each corner; UNDO, CANCEL, DONE.
Alerts	Each fence can alert when someone enters, leaves, or both. When you or a team member crosses it, the phone buzzes, shows a notification, and the fence turns red for a minute.
Breadcrumbs	Dotted trails: blue for you, cyan for each team member. Settings → Map: on/off, length (15 min, 1 h, 4 h), Save my trail (.kml), Clear trails.

9.1 Who sees your points and geofences



Share to ATAK is under Settings → Me → ATAK sharing (off by default). Points and geofences placed in ATAK always show in the app. Breadcrumb trails stay on each phone: they are drawn from the positions everyone already sends.

Figure 8. App users always share with each other; ATAK only when you switch it on.

DELETING

Deleting a point or geofence removes it for everyone, including items other people placed. Settings → Map lists every item and has “Delete all my points & geofences”.

10 · CAPABILITIES TEST (RANGE WALK)

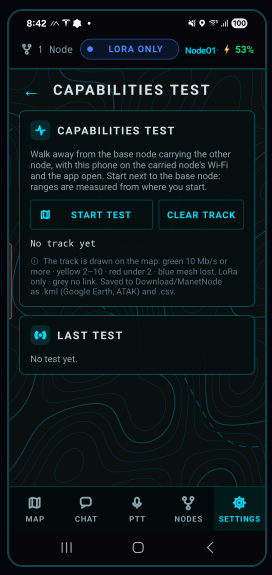
Measures how far each link really reaches where you are: **2.4 GHz**, **HaLow** and **LoRa**. It records a point every 3 seconds with the best link at that spot, draws a coloured track on the map, and saves it.



Figure 9. Walk away from the base node; the track's colour is the link quality at each spot.

1. Leave one node (the base) where it will stand. Carry the other node, with your phone on **the carried node's Wi-Fi**, and the app open.
2. Stand next to the base node. ManetNode → Settings → CAPABILITIES TEST → START TEST. The screen stays on.
3. Walk out until you lose everything, then come back if you like. The page shows the ranges reached so far.
4. STOP & SAVE. The “Last test” card shows the furthest range for 2.4 GHz, HaLow and LoRa, and the Settings menu shows them too.

The track is saved to `Download/ManetNode` as **.kml** (opens in Google Earth and ATAK) and **.csv** (spreadsheet: time, position, link, speed, signal). Points with a GPS accuracy worse than 50 m are skipped.



Settings → Capabilities Test

After a test, the **Last test** card lists the furthest range for 2.4 GHz mesh, HaLow mesh and LoRa, the date, how far you walked, how long it took and how many points were recorded. A link that never showed up says “not seen”.

CLEAR TRACK removes the coloured track from the map; the saved files stay.

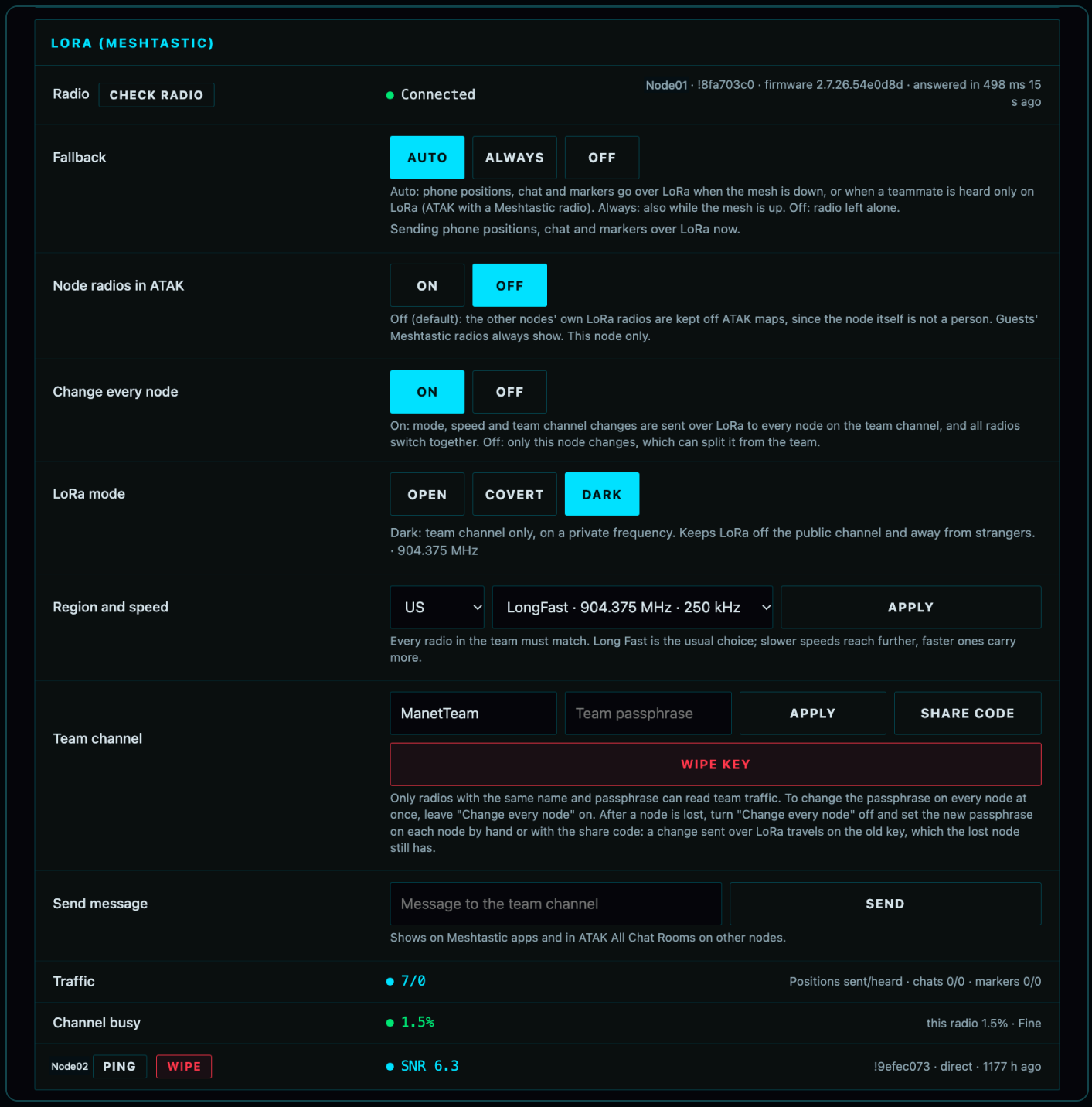
11 · LORA FALLBACK

Each node can have a Meshtastic LoRa radio on USB. LoRa is slow (about 1 kbit/s) but reaches much further than Wi-Fi, so the nodes use it as a backup: **positions, chat and map markers** go over LoRa to nodes and LoRa-only teammates the mesh can't reach. Photos, video, files and voice don't.

FALLBACK	BEHAVIOUR
AUTO	Use LoRa when the mesh is down, or while a LoRa-only teammate is heard. Recommended.
ALWAYS	Also while the mesh is up.
OFF	Leave the radio alone.

LORA MODE	BEHAVIOUR
DARK	Team channel only, on a private frequency worked out from the team name. Keeps LoRa off the public channel and away from strangers. Use this in a mixed team.
COVERT	Team channel + the public channel; your position never goes on the public channel. Public frequency.
OPEN	Team + public channel; public Meshtastic radios see you. Public frequency.

MIXED TEAMS
OPEN and COVERT move the radio to the public LongFast frequency. Radios on a custom layout stop hearing it. Every radio in the team must use the same region, speed and team channel.



Web page → LoRa: radio status, fallback, mode, region and speed, team channel with Share code, a test message, traffic, and each radio heard with its signal.

11.1 Team channel

Only radios with the same team name and passphrase can read team traffic. With **Change every node** ON, a new name or passphrase is sent over LoRa to every node so they all switch together. If a node is lost, turn Change every node OFF and set a new passphrase on each remaining node by hand, so the lost node can't follow.

11.2 Adding a LoRa-only teammate

1. They need a Meshtastic radio paired to their phone, ATAK, and the Meshtastic ATAK plugin.

2. Plugin **LORA** → Team channel → **QR**, or web page → **SHARE CODE**. It holds the team key: only show it to your team.
3. They scan it in the Meshtastic app (Channels → QR), on the same region and speed.
4. Their position and chat appear for everyone on the mesh; yours reach them over LoRa.

11.3 Node radios on the ATAK map

The nodes' own LoRa radios are kept off ATAK maps by default, because the node already shows itself. Guests' Meshtastic radios always show. Switch with “Node radios in ATAK”.

11.4 Chat alerts

Nodes can post battery, mesh and temperature warnings in ATAK's All Chat Rooms. This is OFF by default so the team chat stays clean; the warnings still show on the node's page and in the plugin.

12 · THE MESH: HALOW AND 2.4 GHZ

Nodes link over **HaLow** (sub-GHz, long range, about 15 Mb/s) and, with the USB stick, over **2.4 GHz** (short range, about 30 Mb/s or more). The mesh (batman-adv) picks the best path by itself and switches when a link fades. Phones join the node's own Wi-Fi.

FEATURE	WHAT IT DOES
Link watchdogs	Reboot the node if the HaLow radio stops, or if all HaLow peers vanish for 20 minutes (each at most twice in 6 h); unstick a stuck HaLow rate; move traffic off a weak 2.4 GHz link; check each mesh link.
Phone Internet	**AUTO**: phones use the node's internet when it has one, otherwise their own cellular data. **CELL**: phones always use cellular; the node carries mesh traffic only. Phones switch at their next Wi-Fi reconnect.
Join another mesh	Scan another team's mesh QR to join it, for good or temporarily: a temporary join switches back after the hours you choose. **KEEP** cancels the switch-back.
Share this mesh	Shows the mesh settings as a QR code (includes the mesh password).
Position source	Auto, the node's GPS, or the connected phone: a node with no GPS can take its position from the phone on its Wi-Fi.

RULES THAT KEEP NODES TALKING

- All nodes: same mesh ID, channel, width, key and the 1 MHz primary channel index 1 (the OpenMANET 1.8 default), with HaLow on network `batmesh0`. The self-test checks these.
- RT5370 sticks: use HT20. If a link peers but carries no data, set `rt2800usb nohwcrypt=1` in `/etc/modules.d/rt2800-usb` and reboot.
- Never restart Wi-Fi remotely (`wifi down/up`): it can hang the radios. Reboot instead.

13 · MAINTENANCE, UPDATES AND BACKUPS

13.1 Maintenance mode (white light)

Tells the team “this node is being worked on, don't unplug it or count on it right now”. The light turns **white** by itself while the node is updating, applying settings or running diagnostics. You can also switch it on by hand: ManetNode → Node Settings → **Maintenance mode** → `ON 30 MIN`, the plugin's `SET` tab, or the web page's Maintenance section. It ends after 30 minutes or when you press `OFF`.

13.2 Updating

- **From a computer:** the node installer (3.3) with the newer .ipk.
- **Over the mesh:** update one node, and the others can take the same version from it (web page → Updates from the mesh). **Ask** waits for you to press Update; **Auto** installs by itself. The download must match its checksum.
- **OpenMANET itself:** web page → OpenMANET Updates.

13.3 Self-test and diagnostics

```
ssh root@10.41.0.1 'sh /opt/field-extensions/selftest.sh'

PASS  Python is installed.
PASS  Battery-board connection (I2C bus 1) is on.
PASS  Battery board (PiSugar) answers. Battery 61%.
PASS  Web page and API is running.
PASS  Status light driver is running.
PASS  Phone internet helper is running.
PASS  LoRa service is running.
PASS  Self-recovery is running.
PASS  Button long press: clean shutdown (PiSugar soft-shutdown mode on).
PASS  Power-off step is on (the node turns fully off after a Field Extension shutdown).
PASS  Field Extension page answers at http://10.41.0.1:8787/
PASS  Status light reports maintenance (white light) state.
PASS  HaLow mesh carries mesh traffic (network batmesh0).
PASS  HaLow primary channel index is 1 (matches the other nodes).
RESULT: all checks passed.
```

One line per check: PASS, FAIL (with what to fix) or SKIP (not used on this node). The installer runs it after every install. Web page → Maintenance → **Open diagnostics** collects versions, mesh, HaLow, LoRa, logs and restarts in one page to send to whoever is helping, with no passwords or keys.

13.4 Backups

The installer saves a backup before every install. OpenMANET's own backup (System → Backup) includes the Field Extension's settings, because the package lists them in `/etc/sysupgrade.conf`. To move a node to a new SD card: back up, flash the new card, restore the backup on first boot.

13.5 Clear fleet chat

Deletes the team chat stored on every node and every ManetNode phone, so the next operation starts clean. A node or phone that is off picks it up when it rejoins. ATAK keeps its own chat history.

14 · TROUBLESHOOTING

PROBLEM	TRY
Plugin or app says “not answering”	Check the phone is on a node's Wi-Fi (not cellular-only). In the plugin, long-press the node name and type its address.
Light amber	No other node heard and no LoRa. Check the other nodes are on and within range; reseat the LoRa radio's USB cable.
Light stays red	Open the web page → Node Health and Services to see what failed. A reboot clears most faults.
LoRa “not connected”	Reseat the radio's USB cable. The node retries and power-cycles the USB port by itself.
LoRa connected but nobody heard	All radios need the same region, speed and team channel. Use DARK in a mixed team.
A node hears the mesh but nothing reaches it	Reboot it (PiSugar off, then on). If it keeps happening on the 2.4 GHz link, set the RT5370 fix (chapter 12).
Node won't start after the battery ran flat	Plug in the charger and tap the PiSugar button. Turn auto power-on ON so it starts by itself next time.
Case USB-C charger does nothing	The case socket needs the 5.1 kΩ CC resistors (Figure 4). A USB-A-to-C cable always works.
Battery % jumps or stops near 92 %	Run a battery calibration (chapter 5).
Node changes ask for a password	It's the node's root password, the same as the OpenMANET page.
ATAK won't install the plugin	Use the signed APK that matches your ATAK version (5.7 or 5.8). A developer (debug) build of ATAK needs a debug plugin instead.

15 · REFERENCE

Addresses and ports

WHAT	WHERE
First node	`10.41.0.1` (others get their own 10.41.x.x address)
Field Extension page and API	TCP 8787 · API under `/api/field/`
OpenMANET	TCP 8080, 8081, 8087 (left alone)
ATAK positions (SA)	UDP multicast 239.2.3.1:6969
ATAK chat (GeoChat)	UDP multicast 224.10.10.1:17012
ManetNode points and geofences	UDP multicast 239.2.3.2:6970
Push-to-talk	UDP multicast from the node's comms settings (default 239.41.0.1:38801), Opus audio

On the node

`/opt/field-extensions/`	The program, web page and `selftest.sh`
`/etc/field-extensions/`	Settings (`node.conf` : light pins) and state
`/etc/init.d/fe-*`	The services (fe-web, fe-light, fe-lora, fe-recovery, …)

API (read)

`GET /api/field/summary` (everything on the status page), `/hello` (name, callsign, version; no sign-in), `/nodes` (nodes on the mesh), `/lora` , `/services` , `/card` .

API (write, needs sign-in)

`POST /api/field/…` : `led` , `maintenance` , `power/auto-power-on` , `battery/size` , `battery/calibration` , `battery/safe-shutdown` , `battery/long-press` , `node/identity` , `node/position` , `node/position/source` , `node/reboot` , `node/shutdown` , `node/setup-code` , `node/setup` , `mesh/share` , `mesh/join` , `mesh/join/cancel` , `eud/internet` , `lora/mode` , `lora/profile` , `lora/radio` , `lora/channel` , `lora/node-radios` , `lora/send` , `fleet/clear` , `update/mesh` , `openmanet/update` , `recovery/setting` .

16 · BUILDING FROM SOURCE

FOLDER	BUILD
node-package/	The package's files (`files/`) and opkg control scripts (`CONTROL/`). Pack them into an .ipk with the usual opkg tools.
sd-image/	`build.sh` turns the stock OpenMANET 1.8 image into the ready image. Runs in Docker (arm64 Debian): see its README.
atak-plugin/	ATAK CIV SDK plugin. Put it in the SDK's `samples/` , then `./gradlew assembleCivDebug` . Regular ATAK needs a TAK.gov-signed build.
launcher-apps/android/	ManetNode: `./gradlew assembleDebug` with JDK 17.
launcher-apps/ios, desktop, windows	Simple node launchers for iPhone (Xcode, free Apple ID), Mac (Xcode) and Windows (PowerShell, nothing to install).
node-installer/	The Mac and Windows install scripts.
field-card/	The one-page field card (HTML and PDF).

Third-party software keeps its own license: ATAK CIV SDK (TAK Product Center terms), OpenMANET / OpenWrt (GPL-2.0), Meshtastic protobuf definitions (GPL-3.0). PortableTAK is a separate project.